Your Presentation Title

Your Name

November 8, 2024

Basic Concepts of Navigation Part IV

Mike Carroll GPS SE&I

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Topics

Navigation System Architecture Navigation Aids Velocity Sensors Distance Sensors Position Fix Attitude Sensors Radio Navigation Direction finding Ranging Satellite Navigation GPS

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- *Topics*

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Navigation System Architecture *Navigation Aids* *Velocity Sensors* *Distance Sensors* *Position Fix* *Attitude Sensors* *Radio Navigation* *Direction finding* *Ranging* *Satellite Navigation* *GPS*

- *Topics*
- *Navigation System Architecture*

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- *Velocity Sensors*

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Navigation System Architecture

Primary Sensor INS or GPS Secondary Sensors Navigation Aids Integration / fusion of all sensor inputs via a Kalman filter in a central computer Usually formulated as an error state filter to maximize linearity "central computer" could be embedded in primary navigation sensor

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Navigation Aids

Velocity Sensors Speedometers Magnitude of velocity vector in direction of motion Doppler Velocity Sensors (DVS) Aircraft navigation Senses frequency shift of electronmagnetic beam in three directions Slaved to body coordinates Distance Sensors Odometers Distance along path traveled

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- *Navigation Aids*
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- *Speedometers*

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Navigation Aids

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- *Magnitude of velocity vector in direction of motion*
  - *Magnitude of velocity vector in direction of motion*
- *Doppler Velocity Sensors (DVS)*
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- *Aircraft navigation*
  - *Aircraft navigation*
- *Senses frequency shift of electronmagnetic beam in three directions*
  - *Senses frequency shift of electronmagnetic beam in three directions*
- *Slaved to body coordinates*
  - *Slaved to body coordinates*
- *Distance Sensors*

Navigation Aids

Velocity Sensors Speedometers Magnitude of velocity vector in direction of motion Doppler Velocity Sensors (DVS) Aircraft navigation Senses frequency shift of electronmagnetic beam in three directions Slaved to body coordinates Distance Sensors Odometers Distance along path traveled

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INS Attitude Heading Reference System (AHRS) Three gyros plus vertical accelerometer Possibly magnetic compass input along with magnetic variation tables to correct for gyro drift Some experimentation using multiple GPS antennas widely separated

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Radio Navigation VOR: VHF Omnidirectional Range sensor Receiver senses direction of signal (the radial) from external source and computes distance and bearing to external source location Intersection of two radials yields a position fix DME: Distance Measuring Equipment TACAN: Tactical Air Navigation = VOR + DME for military (more accurate) LORAN−C: Long Range Navigation Time difference of arrival from two sources Yields hyperbolic lines of position Third source needed to intersect the hyperbolas One master station and two secondary stations

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LORAN Hyperbolic LOPs

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- *Navigation Aids*

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- *Satellite Navigation*

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- *Satellite Navigation*
- *Global Positioning System (GPS)*

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- *Six orbital planes, inclined at 55 degrees*

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  - *Six orbital planes, inclined at 55 degrees*
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  $R = c\$\$\text{Delta}\$T\text{toeachofthevisiblesatellites}$

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Navigation Aids: GPS

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Navigation Aids: *GPS*

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- *Navigation Aids*: *GPS*

Navigation Aids: *GPS*

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- *Navigation Aids*: *GPS*
- *Known*

Navigation Aids: *GPS*

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-
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- *Navigation Aids*: *GPS*
- *Known*
- *Unknown*

Navigation Aids: *GPS*

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-
- *Navigation Aids*: *GPS*
- *Known*
- *Unknown*
- *Pseudoranges*:

Navigation Aids: *GPS*

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- *Navigation Aids*: *GPS*
- *Known*
- *Unknown*
- *Pseudoranges*:
- *Transmitter*

Navigation Aids: *GPS*

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- *Navigation Aids*: *GPS*
- *Known*
- *Unknown*
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- *Transmitter*
- *Locations*:

Navigation Aids: *GPS*

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Navigation Aids: *GPS*

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- *User Location*:
- *Solution*

Navigation Aids: *GPS*

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- *Unknown*
- *Pseudoranges*:
- *Transmitter*
- *Locations*:
- *User Location*:
- *Solution*
- *Start with initial guess*:

Navigation Aids: GPS

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- *Iteratively solve for new estimates*,

Navigation Aids: GPS

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Navigation Aids: GPS

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Navigation Aids: GPS

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- *Navigation Aids*: *GPS*
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Navigation Aids: *GPS*

- *Navigation Aids*: *GPS*
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Navigation Aids: GPS

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Navigation Aids: *GPS*

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Navigation Aids: GPS

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Navigation Aids: GPS

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Navigation Aids: GPS

Navigation Aids: *GPS*

- *Navigation Aids*: *GPS*

Navigation Aids: *GPS*

- *Navigation Aids*: *GPS*
- *If*

Navigation Aids: *GPS*

- *Navigation Aids*: *GPS*
- *If*
- *then*

Navigation Aids: *GPS*

- *Navigation Aids*: *GPS*
- *If*
- *then*
- $GDOP =$

Navigation Aids: *GPS*

- *Navigation Aids*: *GPS*
- *If*
- *then*
- $GDOP =$
- $TDOP =$

Navigation Aids: *GPS*

- *Navigation Aids*: *GPS*
- *If*
- *then*
- $GDOP =$
- $TDOP =$
- $HDOP =$

Navigation Aids: *GPS*

- *Navigation Aids*: *GPS*
- *If*
- *then*
- $GDOP =$
- $TDOP =$
- $HDOP =$
- $VDOP =$

Navigation Aids: *GPS*

- *Navigation Aids*: *GPS*
- *If*
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- $GDOP =$
- $TDOP =$
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- $VDOP =$
- $PDOP =$

Navigation Aids: GPS

- *Navigation Aids: GPS*
- *If*
- *then*
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- $HDOP =$
- $VDOP =$
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- *Note: GDOPS are related*

Navigation Aids: GPS

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Navigation Aids: GPS

-

Navigation Aids: *GPS*

-
-
- *Navigation Aids*: *GPS*

Navigation Aids: GPS

-
-
- *Navigation Aids: GPS*
- *Good DOP*

Navigation Aids: GPS

-
-
- *Navigation Aids: GPS*
- *Good DOP*
- *Bad DOP*

Navigation Aids: GPS

-
-
- *Navigation Aids: GPS*
- *Good DOP*
- *Bad DOP*

Navigation Aids

Velocity is calculated using range rate GPS Time: Composite clock of all SV clocks in the constellation Steered to within 1 microsecond of UTC, disregarding leap seconds

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Navigation Aids: GPS

Error Sources Satellite Clock and Ephemeris Errors Ionospheric Delays Tropospheric Delays Receiver Noise and Resolution Multipath Dual−Frequency Receivers and Differential GPS can eliminate much of the Ionospheric delay

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- *Error Sources*

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- *Tropospheric Delays*
  - *Tropospheric Delays*
- *Receiver Noise and Resolution*
  - *Receiver Noise and Resolution*
- *Multipath*

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  - *Receiver Noise and Resolution*
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